Free-fall lab

antocciphysics

Name: _____ **Block:** _____ **Mastery Rating:** (teacher only) _____

LEARNING TARGET Estimate g from measured motion and discuss measurement limits.

SUCCESS CRITERIA - I can...

1. Keep supplied examples separate from measurements.
2. Find acceleration from a velocity-time slope.
3. Use repeated trials and explain uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “slope” and “trial” to explain your model or result. Sentence starters are optional.

slope: vertical change divided by horizontal change on a graph

trial: one run or repetition of a test

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

A supplied gravity model has $Gm_1m_2=75$ N·m² and separation 5 m.
Find the force and its direction.

Word help: separation = distance between the specified objects or centers

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $a = \Delta v/\Delta t$; $v_f = v_i + a\,\Delta t$.
Velocity-time slope estimates acceleration. Downward is positive. Use the approved apparatus; distinguish supplied from measured data.

Worked reasoning: Supplied trial values 8, 10, 12 m/s² have mean 10 m/s² and range 8-12. The spread describes variation; averaging does not remove every systematic error.

Your turn: Plot the three values on a number line and mark the mean.
Name an error that could shift all trials together.

Sentence starter: The step I would change is ... . The evidence is ... .

Trial	Estimated g (m/s²)
1	8
2	10
3	12

Marzano check - circle: ① ② ③

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Continue down each column.

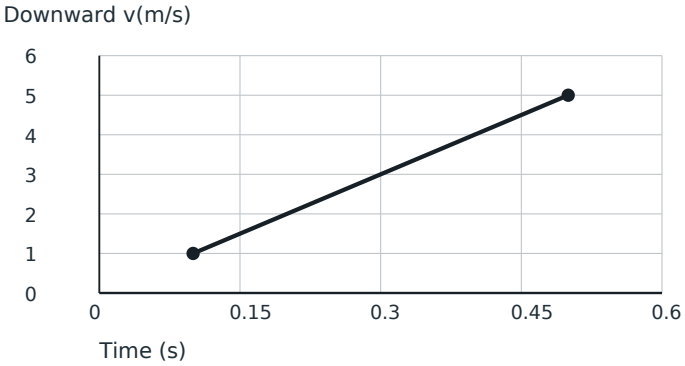
Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Use the supplied velocity-time graph to estimate acceleration.
Show the change in velocity and elapsed time.

Word help: estimate = find an approximate value

Sentence starter: I found … [unit]. This value means … in this situation.

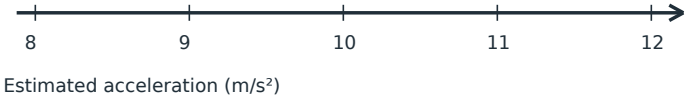


Marzano check - circle: 1 2 3

04 Apply the physics

Three video measurements of a falling ball give accelerations of 9, 10 and 11 m/s².
Find the mean and the lowest-to-highest interval.
Mark all three measurements and the mean on the number line.

Sentence starter: I found … [unit]. This value means … in this situation.



Estimated acceleration (m/s²)

Marzano check - circle: 1 2 3

05 Explain your reasoning

Plan three repeated trials.
Name a controlled condition, a measurement you will record and a reason not to include a badly pushed release without comment.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A box at rest has 18 N right and 4 N left.
Find the additional horizontal force needed for balance and show the sum.
Draw the forces on the chosen object and label their sources.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

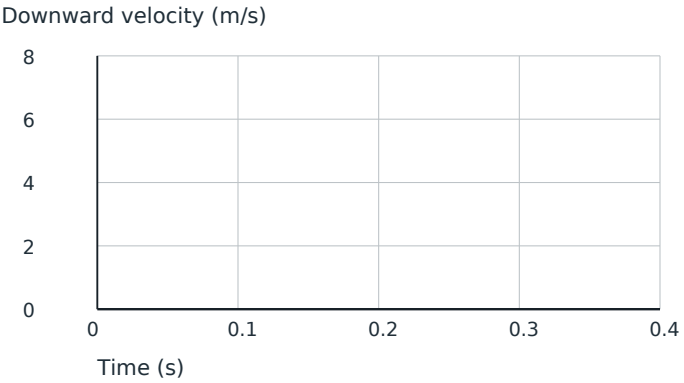
Add and label force arrows.

Marzano check - circle: ① ② ③

07 Exit check

A motion sensor records a falling ball speeding up from 4 to 6 m/s over 0.4 s. Take downward as positive.
Draw the velocity-time segment and calculate its slope.
What acceleration does this measurement imply?

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

08 STRETCH: test the model

Two groups have values 9, 10, 11 and 5, 10, 15 m/s².
Compare means and repeatability without declaring either perfectly accurate.

Sentence starter: Both … . They differ in … because … .

Group / trial	Values (m/s²)	Mean	Spread
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③